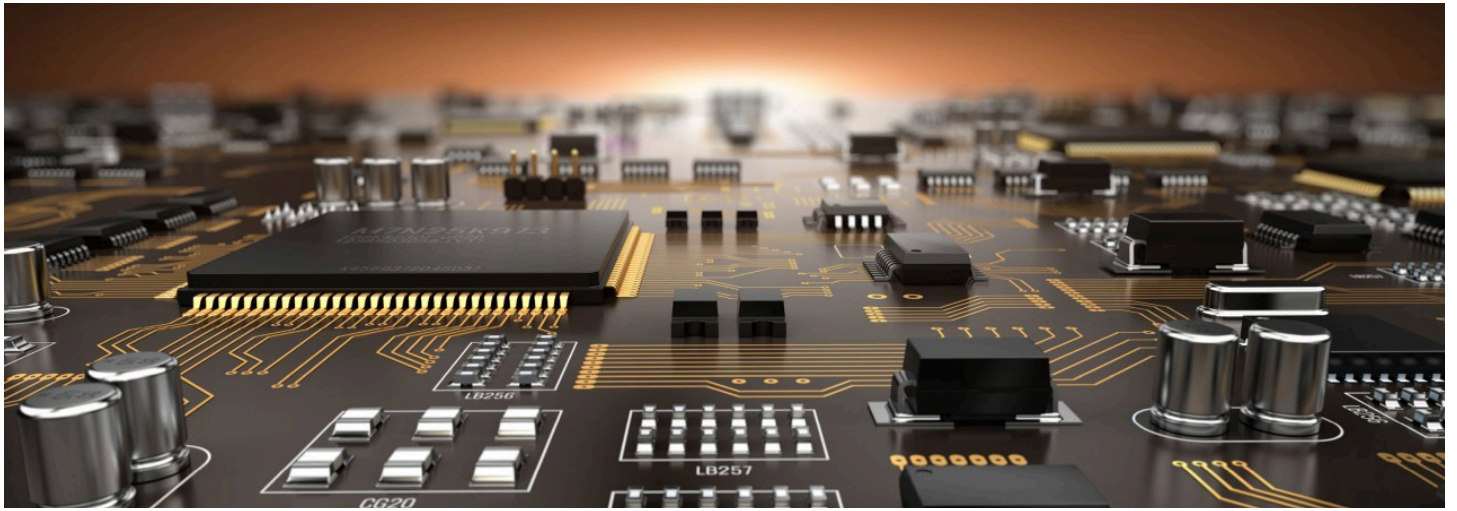




Analysis and Development of BMS in Energy Storage Batteries



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We have earlier talked a lot about electric vehicles and their Battery Management Systems. Currently, the vehicles depend on the grid power supply but as people will buy more and more electric vehicles, the grid will need some help with energy storage infrastructure.

To manage the batteries in energy storage devices a BMS is needed. Today, we will learn about this battery management system in detail.

What is a Battery Management System (BMS)?

The Battery Management System or BMS is responsible to intelligently manage each battery unit. It prevents overcharge and discharge. BMS is important to extend battery service life and monitor the battery's state.

Composition of Battery Management System

BMU, current sensor, insulation monitoring, and BCU are crucial components in the power battery system.

BMU, or Battery Management Unit, gathers battery current, voltage, temperature, and related data. It then transmits this information to the BCU for control.

The BCU, or Brake Control Unit, manages the power battery system and interacts with other control units. It receives various data, performs counting, calculation, and analysis, calculates SOC/F/H, controls fan or heating, and communicates with the entire vehicle and charger.

How the Battery Management System Works

The Battery Management System is a system that monitors and manages battery status. Its primary role is to guarantee that the battery operates safely and reliably while also extending its service life. Its functioning principles mostly contain the following four items:

- 1. Monitoring:** The BMS will monitor the battery's status, including voltage, current, and temperature. Monitoring these metrics allows us to better understand the battery's health, performance, and safety.
- 2. Protection:** BMS safeguards the battery against overcharge, over-discharge, short circuit, and other hazards. When the battery's voltage or temperature exceeds the permissible range, BMS will take appropriate precautionary steps to prevent damage or risk to the battery.
- 3. Equalization:** The BMS balances the voltage of various monomers in the battery pack to minimize battery life loss or other issues caused by high voltage differences. This balance helps to keep the battery consistent and extends the battery pack's service life.
- 4. Communication:** The BMS allows for remote monitoring and control of battery management by interacting with the vehicle control system. This connection allows the vehicle management system to better coordinate and control the battery's operational state.

The Main Functions of Energy Storage BMS